Revisit quotient factorisation, operator Carleson extension, and signed-score boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

This note executes the direct source-action successor selected after the R-117 loop audit. Its purpose is to decide whether a visit-count-free operator-valued version of the R-068 centered-form estimate exists, and to state exactly what a complete adapted owner must satisfy before that estimate can be paid by one terminal sextic rather than by a visit square function.

What is proved. For a finite visit space, a selfadjoint quadratic form descends to the endpoint quotient if and only if it annihilates the synthesis kernel, equivalently if and only if it has the form L^*BL. The R-068 centered-form theorem extends without a visit-count loss to spatially constant selfadjoint operator families whose absolute spectral variation is bounded. If those operators factor through the endpoint synthesis with a uniformly bounded endpoint family, the analytic estimate uses the endpoint field rather than the visit square function.
Every scalar Gaussian polynomial has a canonical decomposition into its zero chaos, first chaos, and one double divergence. Thus a pure double-divergence representation exists exactly when the zero and first chaoses vanish. An exact two-visit Hermite quotient has mean $-\epsilon^2$ and a unique sign-changing scalar preimage, disproving a universal positive-semidefinite random-W factorisation.

What is not proved. R-104 and R-116 prove quotient descent for the recombined endpoint packet in their fixed-environment scope; they do not prove that an isolated Hessian, trace, current, or forest owner descends. No current result verifies the zero/first-chaos cancellation, vertical basicness, random-W forest compatibility, or spatial multiplier bound for the full adapted production cluster. The progressive/revisit extension, one-use aggregation, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson estimate, removals, interacting measure, Sector-A closure, and tier promotion remain open.

The analytic theorem is finite-cutoff and uses the fixed $L = 16$ torus and the sharp-cube maximal bounds already established in R-068/A12. The Gaussian range theorem is finite dimensional and polynomial. The Hermite fixture is an abstract contraction-connected output diagnostic, not an independently embedded A1 production root.

2. The finite-visit quotient theorem

Let V and H be finite-dimensional real Hilbert spaces, let $L : V \to H$ be linear, and let $K = K^* : V \to V$. Write

$$q_K(v) = \langle v, Kv\rangle. \tag{2.1}$$

Theorem 2.1 (quadratic quotient factorisation)

The following statements are equivalent.

1. q_K is constant on every fibre $v + \ker L$.
2. $\ker L \subseteq \ker K$.
3. There is a selfadjoint B on $\operatorname{Ran} L$ such that

$$K = L^* B L. \tag{2.2}$$

For the Moore–Penrose inverse $L^\dagger$, the canonical quotient operator is

$$B = (L^\dagger)^* K L^\dagger \quad \text{on } \operatorname{Ran} L. \tag{2.3}$$

Moreover, for $C \geq 0$, the form bound

$$|\langle v, Kv \rangle| \leq C \|Lv\|^2 \quad (v \in V) \tag{2.4}$$

is equivalent to a factorisation (2.2) with $\|B\| \leq C$, equivalently

$$-CL^*L \leq K \leq CL^*L. \tag{2.5}$$

Proof. If $k \in \ker L$, fibre invariance applied to $v + tk$ says that

$$2t\langle v, Kk \rangle + t^2 \langle k, Kk \rangle = 0 \quad \text{for every } v \in V,\ t \in \mathbb{R}. \tag{2.6}$$

The linear coefficient gives $Kk = 0$. Conversely, $Kk = 0$ makes (2.6) zero. When K kills $\ker L$, the form

$$\langle Lv, BLw \rangle := \langle v, Kw \rangle \tag{2.7}$$

is well defined and selfadjoint on $\operatorname{Ran} L$, and gives (2.2). Formula (2.3) is the same construction in the orthogonal representative $(\ker L)^\perp$. Finally (2.4) defines a bounded selfadjoint form on $\operatorname{Ran} L$ with norm at most C; the finite-dimensional representation theorem gives B. The converse and (2.5) are immediate. □

It is not enough to check only $q_K(k) = 0$ on $\ker L$ for an indefinite form; the mixed polarisation in (2.6) is load-bearing.

3. Fixed linear endpoints and the curvature boundary

Let $\Phi : H \to \mathbb{R}$ be C^2, let $A_0 \in H$, and suppose that the visit dependence is genuinely through one fixed affine synthesis:

$$F(v) = \Phi(A_0 + Lv). \tag{3.1}$$

Then

$$DF(v) = L^* D\Phi(A_0 + Lv), \qquad D^2F(v) = L^* D^2\Phi(A_0 + Lv) L. \tag{3.2}$$

Thus the full endpoint Hessian annihilates every closed revisit cycle. This is the correct algebra behind the R-116 endpoint quotient when all objects being differentiated are functions of the same fixed endpoint data.

The statement does not apply termwise to an adapted owner decomposition. If $E : V \to H$ is nonlinear, then

$$D^2(\Phi \circ E) = (DE)^* D^2\Phi(E)DE + D\Phi(E)\, D^2E. \tag{3.3}$$

The curvature term contains the current Hessian, trace, and forest companions. If coefficients also depend on a vertical variable, their first and second vertical derivatives reappear as the DW, D^2W forest. Therefore an isolated owner factors only after one proves its full vertical basicness, or after the complete signed owner cancellation proves the same fact. R-104/R-116 permit mean and covariance mass to exchange between owners, so endpoint invariance of their recombined scalar packet does not by itself establish termwise (2.2).

4. Operator-valued centered-form theorem

Let $\mathcal{H}$ be a real Hilbert space and let $a \in H^2(\mathbb{T}^3;\mathcal{H})$. Let $K_r = K_r^*$ be a finite family of spatially constant operators satisfying the absolute spectral-variation bound

$$\sum_r |K_r| \le K_0 \mathrm{I}_{\mathcal{H}}. \tag{4.1}$$

With the R-068 sharp-cube projections, put

$$q_r(x) = \langle a(x), K_r a(x)\rangle_{\mathcal{H}}, \qquad \Pi_K(a) = \sum_r S_{r-1} q_r, \tag{4.2}$$

$$X = \|a\|^2_{H^2(\mathbb{T}^3;\mathcal{H})}, \qquad Y = \||a|_{\mathcal{H}}\|^6_{L^6}. \tag{4.3}$$

Theorem 4.1 (visit-count-free operator R-068 extension)

There are constants depending on the fixed torus and sharp-cube system, but not on the number of operators, such that

$$\|\Pi_K(a)\|_2 \le C_0 K_0 Y^{1/3}, \qquad \|\Pi_K(a)\|_{H^2} \le C_2 K_0 X. \tag{4.4}$$

For $0 < \kappa < 1$, with $\theta = (1+\kappa)/2$,

$$\boxed{\|\Pi_K(a)\|_{H^{1+\kappa}} \le C_\kappa K_0 X^{(1+\kappa)/2} Y^{(1-\kappa)/6}.} \tag{4.5}$$

Consequently, if $M = \|Q\|_{H^{-1-\kappa}}$, then for every $\eta, \zeta > 0$,

$$\begin{aligned} |\langle Q, \Pi_K(a)\rangle| &\le \eta X + \zeta Y \\ &\quad + C_\kappa K_0^{3/(1-\kappa)} \eta^{-3(1+\kappa)/[2(1-\kappa)]} \zeta^{-1/2} M^{3/(1-\kappa)}. \end{aligned} \tag{4.6}$$

At the production value $\kappa = 1/10$, the powers of $(X, Y, \text{Young gap})$ are $(11/20, 3/20, 3/10)$, and the random moment and K_0 powers are $10/3$; the negative η and ζ powers are $11/6$ and $1/2$.

Proof. For every $u, v \in \mathcal{H}$, polar decomposition and Cauchy–Schwarz in the index r give

$$\sum_r |\langle u, K_r v\rangle| \le \left\langle u, \sum_r |K_r| u \right\rangle^{1/2} \left\langle v, \sum_r |K_r| v \right\rangle^{1/2} \le K_0 \|u\| \|v\|. \tag{4.7}$$

For $g \in L^2$, selfadjointness of S_r, (4.7), the sharp-cube maximal theorem at $3/2$, and finite volume give

$$|\langle \Pi_K(a), g\rangle| \le K_0 \int |a|^2 S_*|g| \le CK_0\||a|^2\|_3\|g\|_2, \tag{4.8}$$

which proves the first bound in (4.4). For $|\alpha| \le 2$, differentiate before dualising:

$$\partial^\alpha q_r = \sum_{\beta\le\alpha} c_{\alpha,\beta}\langle \partial^\beta a, K_r\partial^{\alpha-\beta} a\rangle. \tag{4.9}$$

Equation (4.7), the L^2 sharp-cube maximal theorem, and the Hilbert-valued three-dimensional H^2 algebra estimate prove the second bound in (4.4). This differentiated duality is essential; the generally unjustified shortcut $\sum_r \|q_r\|_{H^2} \lesssim X$ is not used. Interpolation gives (4.5). Three-factor weighted Young with gap $(1-\kappa)/3$ gives (4.6). □

Spatially varying $K_r(x)$ require a uniform multiplier bound at the same Sobolev order. Uniform operator norm alone is insufficient: a constant visit multiplied by e^{iMx} has fixed pointwise operator norm and an H^2 norm growing like M^2.

5. Terminal ownership and the opposite-visit test

Suppose $L : \mathcal{H}_{\text{visit}} \to \mathcal{H}_{\text{end}}$ is the endpoint synthesis and

$$K_r = L^* B_r L, \qquad \sum_r |B_r| \le B_0 \mathrm{I}. \tag{5.1}$$

Then $q_r(a) = \langle La, B_r La\rangle$, so Theorem 4.1 applies directly to the terminal aggregate $z = La$. This is the sufficient bridge from the analytic visit theorem to a terminal-sextic one-use payment. It still requires a cutoff-uniform source estimate for $\|z\|_{H^2}^2$ and the declared terminal L^6 bound.

The condition is sharp at the quotient level. With two visits,

$$L = (1, 1), \qquad a = (f, -f), \qquad K_{\text{diag}} = \mathrm{I}_2, \tag{5.2}$$

one has

$$La = 0, \qquad \langle a, K_{\text{diag}} a\rangle = 2|f|^2, \qquad Y_{\text{visit}} = 8\|f\|_6^6. \tag{5.3}$$

Thus a diagonal visit square cannot be paid by the zero terminal sextic. The coherent endpoint matrix L^*L does kill $(f, -f)$. Likewise m coherent visits $f/\sqrt{m}$ have fixed visit source energy and terminal square $m|f|^2$; a uniform source theorem therefore also needs a uniform synthesis bound. These fixtures do not refute Theorem 4.1, whose Y is explicitly the visit square-function sextic, or the complete signed action.

6. Canonical Gaussian double-divergence range

Let $G \sim N(0, \mathrm{I}_n)$ and let a scalar polynomial have Wiener-chaos expansion

$$F = \sum_{k=0}^{d} F_k. \tag{6.1}$$

For a symmetric polynomial matrix W, use

$$\delta^2 W = G^T W G - \operatorname{Tr} W - 2G \cdot \operatorname{Div} W + \operatorname{Div}^2 W. \tag{6.2}$$

Theorem 6.1 (canonical second-divergence decomposition)

Define

$$W_{\mathrm{can}} = 2\sum_{k=2}^{d} \frac{D^2F_k}{k(k-1)}. \tag{6.3}$$

Then

$$\boxed{F = \mathbb{E}F + \langle \mathbb{E}[GF], G\rangle + \frac{1}{2}\delta^2 W_{\mathrm{can}}.} \tag{6.4}$$

Hence $F = \delta^2 W/2$ for some polynomial symmetric W if and only if

$$\mathbb{E}F = 0, \qquad \mathbb{E}[GF] = 0. \tag{6.5}$$

The right inverse (6.3) is canonical; in dimension greater than one it need not be the only tensor preimage. In the scalar one-dimensional polynomial case it is unique.

Proof. On the kth homogeneous chaos,

$$\delta^2 D^2 F_k = k(k-1)F_k. \tag{6.6}$$

The range of δ^2 contains no chaos of degree zero or one. Summing (6.6) proves (6.4)–(6.5). In one dimension $\delta^2 H_k = H_{k+2}$, which is injective on polynomials. □

This theorem turns the proposed production identification into two explicit low-chaos tests before any positivity or exponential estimate is attempted.

7. Exact score split and the two-visit mean debt

Let $R : \mathbb{R}^n \to \mathbb{R}^m$ be polynomial, let $x_0 = b + AG$, and set

$$F = \langle x_0, R\rangle + \frac{1}{2}\|R\|^2 - \frac{1}{2}\delta\tau. \tag{7.1}$$

For any symmetric polynomial W, define

$$\tau_W = \operatorname{Tr} W + 2G\cdot \operatorname{Div} W - \operatorname{Div}^2 W, \qquad N_W = \langle x_0, R\rangle + \frac{1}{2}\|R\|^2 - \frac{1}{2}G^T W G. \tag{7.2}$$

Then the pointwise identity

$$\boxed{F = \frac{1}{2}\delta^2 W + N_W + \frac{1}{2}(\tau_W - \delta\tau)} \tag{7.3}$$

holds without an estimate. Thus trace/forest compatibility $\delta\tau = \tau_W$ and a favourable Schur remainder N_W are distinct proof obligations. Neither follows from static positivity of W.

The smallest same-root diagnostic makes the obstruction exact. Let G now be scalar standard Gaussian, let H_k be probabilists' Hermite polynomials, let v be a unit output vector, and for $s = 0, 1/2, 1$ set

$$x_s = \epsilon v\{aG + sH_2(G)\}, \qquad t_s = \epsilon^2(a + 2sG)^2. \tag{7.4}$$

The two R-116 visit packets telescope to

$$
\begin{aligned}
F_{2\mathrm{v}} &= \frac{1}{2}(\|x_1\|^2 - \|x_0\|^2) - \frac{1}{2}(t_1 - t_0) \\
&= \epsilon^2 \left(aH_3 + \frac{1}{2}H_4 - 1 \right).
\end{aligned}
\tag{7.5}
$$

Therefore

$$
\mathbb{E}F_{2\mathrm{v}} = -\epsilon^2, \qquad \mathbb{E}[GF_{2\mathrm{v}}] = 0. \tag{7.6}
$$

The centered part has the unique scalar polynomial preimage

$$
W_{\mathrm{can}} = \epsilon^2(H_2 + 2aH_1) = \epsilon^2\{(G+a)^2 - (a^2+1)\}, \tag{7.7}
$$

$$
F_{2\mathrm{v}} + \epsilon^2 = \frac{1}{2}\delta^2 W_{\mathrm{can}}. \tag{7.8}
$$

The value of (7.7) at $G = -a$ is $-\epsilon^2(a^2+1)$, while it is positive for sufficiently large $|G|$. Thus the exact preimage is signed, and the remaining negative mean cannot be a double divergence. This is the exact authority for `NG-2026-07-28-A13-UNIVERSAL-PSD-RANDOM-W-DOUBLE-DIVERGENCE` and strengthens the mean-debt diagnosis behind `NG-2026-07-28-A13-UNIVERSAL-NONLINEAR-TANGENT-SQUARE-FIRST-NORMALIZER`. It does not produce a legal A1 counterexample.

8. Cartan homotopy does not remove the current defect

R-102 gives, for the isolated complete-L rational current one-form ω, the exact active/kernel fixture

$$
\partial_y \omega_x - \partial_x \omega_y = -\frac{40}{729} \neq 0. \tag{8.1}
$$

On a star-shaped chart the exact homotopy split is

$$
\omega = \mathrm{d}\Phi + \kappa, \qquad \Phi(z) = \int_0^1 \omega(tz)\cdot z\,\mathrm{d}t, \tag{8.2}
$$

$$
\kappa_i(z) = \sum_j z_j \int_0^1 t(\partial_j\omega_i - \partial_i\omega_j)(tz)\,\mathrm{d}t. \tag{8.3}
$$

Since $\mathrm{d}\kappa = \mathrm{d}\omega$, (8.1) implies $\kappa \not\equiv 0$. The current subowner is therefore not itself a scalar endpoint Hessian. R-102's fixture omits the endpoint square, trace, forest, low, and other rows, so it does not refute a factorisation of a future fully recombined cluster. It proves that such a result must use those companions, not silently discard the Cartan remainder.

9. Exact production successor

The new route is now a finite sequence of falsifiable checks on the smallest legal contraction-closed root cluster.

1. After the frozen affine determinant absorbs its declared first chaos, compute the zero and first Wiener chaoses of the complete adapted residual. Failure of either equality in (6.5) identifies the exact mean or linear debt.

2. If both vanish, construct W_{can} by (6.3), rather than guessing a positive covariance weight.

3. Match the actual heat, trace, mixed baseline, rational recovery, and R-063 forest against $\tau_{W_{\mathrm{can}}}$ in (7.3), retaining every owner once.

4. Prove fibre invariance, equivalently vertical annihilation, for the recombined coefficient. Termwise endpoint descent is not assumed.

5. Prove the spectral-variation and spatial multiplier hypotheses of Theorem 4.1 after endpoint factorisation, together with a uniform synthesis bound from the R-093 source Hilbert space.

6. Only then apply (4.6) once over the directed source union and use the centered determinant cost. No shellwise or visitwise Young inequality is permitted.

This replaces the vague request for a positive complete matrix by three sharp tests: low-chaos cancellation, quotient basicness, and operator variation.

10. Executable verification

The primary exact SymPy implementation derives all quotient, exponent, opposite-visit, Hermite, preimage, and score-split identities symbolically. It passes 33/33 assertions. A non-importing standard-library implementation reconstructs the matrix algebra, Hermite recurrence, Gaussian moments, subdivision cancellation, and Cartan homotopy with exact rational arithmetic; it passes 27/27 assertions. The integrated verifier reruns both children, checks deterministic stored artefacts, pins all authorities, sources, note, PDF, manifest, and no-overclaim flags, and rejects a stale package.

The two deterministic run artefacts are stored under the claim's `runs/` directory with the declared primary and independent package slugs. Their exact paths and hashes are pinned by the package manifest.

11. Failed routes and reusable conclusions

Route	Verdict	Reusable conclusion
Diagonal visit square	terminal-owner failure	Opposite visits are invisible at the endpoint but retain positive visit sextic mass.
Universal PSD coefficient	failed	The exact Hermite preimage changes sign and a negative zero-chaos debt remains.
Isolated current primitive	failed	R-102's nonzero curl survives as the Cartan homotopy term.
Quotient algebra alone	boundary	It proves a criterion, not the adapted vertical cancellation or multiplier estimate.
Operator R-068 extension	advanced	Spectral variation removes visit count. Exact fibre descent of the recombined form requires quotient factorisation; the termwise condition (5.1) is sufficient, not proved necessary.

12. Devil's-advocate

1. **Objection: R-116 endpoint descent already proves every owner factors. UPHELD as false.** It proves the recombined fixed-environment endpoint identity. Mean, covariance, trace, and forest owners may exchange.

2. **Objection: kernel inclusion follows from checking $q_K(k) = 0$. UPHELD as false for indefinite forms.** The mixed polarisation in (2.6) must vanish for every base point.

3. **Objection: the operator estimate pays the terminal sextic. UPHELD as unsupported without (5.1).** Theorem 4.1 initially sees the visit square-function sextic. Opposite visits separate the two quantities.

4. **Objection: the Young remainder remains linear in K_0. UPHELD as false.** The three-factor Young step raises both K_0 and M to $3/(1-\kappa)$, equal to 10/3 at $\kappa = 1/10$.

5. **Objection: exact double-divergence centering makes the coefficient PSD. UPHELD as false.** Formula (7.7) is the unique scalar preimage and changes sign.

6. **Objection: the negative Hermite mean disproves the physical A1 cluster. DISMISSED.** It is an abstract connected diagnostic. A larger physical contraction-closed cluster may supply a cancelling mean owner.

7. **Objection: Cartan homotopy integrates away the curl. UPHELD as false.** Equation (8.3) retains the curvature; its exterior derivative is the original nonzero curl.

8. **Objection: a pointwise bounded random coefficient preserves the H^2 estimate. UPHELD as false.** Spatial multiplier regularity and Gaussian DW, D^2W forest compatibility are independent obligations.

13. Result footer

```
Result ID: A13-CLASSII-REVISIT-QUOTIENT-OPERATOR-CARLESON-
  SIGNED-SCORE-BOUNDARY (R-118).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: Theorems 2.1, 4.1, and 6.1; fixed-linear endpoint
  chain rule (3.2); terminal factorisation boundary (5.1)-(5.3);
  exact score split (7.3); Hermite mean/sign boundary (7.4)-(7.8);
  and Cartan homotopy boundary (8.1)-(8.3).
Scope: finite visit spaces; finite operator families; spatially constant
  coefficients for Theorem 4.1; fixed L=16 sharp cubes; finite-dimensional
  polynomial Gaussian range theorem; abstract fixtures are not A1 roots.
Dependencies: R-063, R-068, R-093, R-102, R-104, R-107, R-110,
  R-116, R-117.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command:
  E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_revisit_quotient_operator_carleson_signed_score_
  boundary_verify.py
Expected output: Integrated R-118 PASS; primary 33/33 and independent
  27/27 children PASS; all authority, source, note, PDF, manifest,
  and stored-result pins valid.
Falsification gate: any failed quotient equivalence, spectral exponent,
  opposite-visit, Hermite, double-divergence, score-split, Cartan,
  child/source/note/PDF/manifest assertion, or use beyond stated scope.
Tier before / after: T4 / T4.
No-overclaim statement: The universal PSD random-W representation is false,
  but a production-specific positive complete-cluster representation remains
  open. No complete adapted A1 packet, uniform endpoint multiplier,
  progressive/revisit H_A,
  one-use aggregation, OVERLAP_src, Nelson, removal, interacting measure,
  Sector-A closure, or T5-T7 promotion is proved.
Next required action: Compute the zero/first chaoses and canonical W of the
  smallest legal adapted cluster, then prove its complete trace/forest
  compatibility, vertical basicness, and endpoint operator variation.
```